

DI Weather Station Real-Life Scan

Final Operational Report

Location: PA 19341 | Observation Window: ~2:23 PM-5:00 PM ET

Observer	Fixed ground visual observer facing east
Scan Type	Real-time convective nowcast assessment
Data Policy	Low-confidence [REDACTED WEATHER SOURCE] humidity input purged; AccuWeather-weighted surface moisture retained.
Final Status	SCAN ACTIVE - ALERT NOT TRIGGERED

SCAN ACTIVE - ALERT NOT TRIGGERED

1. Environmental Snapshot

Variable	Operational Value
Temperature	~73-74°F
Humidity	~62%
Dew Point	56-58°F
Wind	NW/WNW 9-10 mph
Gusts	13-14 mph
Visibility	5-10 mi
Ceiling	9,000-11,500 ft
Pressure	~29.9 inHg rising

Interpretation

Environment supported:

- scattered convection,
- moisture transport,
- and intermittent precipitation development,

but also contained:

- stabilizing pressure tendencies,

- elevated cloud bases,
- and fragmented organization.

2. Radar / Regional Structure

Radar Characteristics

Observed:

- broken frontal precipitation band,
- embedded convective cells,
- fragmented reflectivity structure,
- uneven precipitation continuity,
- no dominant mesoscale core approaching 19341.

Motion

Estimated:

- east / ENE progression,
- moderate forward motion,
- intermittent regeneration.

Interpretation

Regional precipitation support existed, but organization remained inconsistent and spatially uneven.

3. Webcam / Spatial Verification Layer

Regional webcams showed:

- active precipitation in portions of the broader region,
- variable cloud morphology,
- differing local sky states within short distances.

This supported:

mesoscale fragmentation and localized precipitation behavior

Operational implication:

- rain somewhere nearby became increasingly likely,
- but direct impact certainty over the observer point remained unresolved.

4. Visual Convective Analysis

Positive Indicators

Observed:

- cauliflower texture,
- vertical uplift,
- sunlit tower edges,

- temporary anchoring,
- leaf/wind mechanical mixing.

These confirmed:

active convective energy transfer was occurring

Limiting Indicators

Observed:

- tower weakening within ~10 minutes,
- weak regeneration,
- no mature anvil,
- no persistent rain shaft,
- no strong base lowering,
- no sustained darkening,
- rising pressure trend.

These reduced confidence in:

- sustained thunderstorm maturation,
- organized intensification,
- or severe continuation over 19341.

5. Convective Classification

Final Mode

**Weak-to-moderate fragmented convection
embedded within a broader frontal precipitation regime**

Not classified as:

- organized severe convection,
- supercellular structure,
- or mature sustained thunderstorm development.

6. Probability Assessment

Event	Final Estimate
Any drops / brief shower	65-75%
Measurable rain at observer	60-70%
Sustained rain (>15-20 min)	40-55%
Thunder nearby	35-45%
Direct thunder over observer	20-35%
Severe local impact	Very low

7. Sustainment vs Collapse

Final Determination

System favored partial fragmentation over sustained local intensification.

Primary reasons:

- rising pressure,
- inconsistent tower persistence,
- elevated ceilings,
- weak consolidation,
- fragmented radar morphology,
- uneven regional moisture realization.

8. Radar Lag / Model Disagreement

Disagreement between:

- [REDACTED WEATHER SOURCE],
- AccuWeather,
- radar imagery,
- webcams,
- and direct visual observation

was attributable to:

- fragmented convection,
- rapidly evolving mesoscale structure,
- probabilistic smoothing,
- and spatial precipitation inconsistency.

Key operational principle:

regional precipitation existence
does not guarantee direct local realization

9. Observer Contribution Assessment

High-value observer inputs included:

- repeated timed sky photographs,
- tower evolution tracking,
- leaf/wind observations,
- webcam cross-validation,
- pressure incorporation,
- radar comparison,
- source-confidence filtering.

Observer methodology materially improved:

- mesoscale interpretation,
- spatial continuity assessment,
- and short-term local confidence calibration.

10. Final Operational Decision

DI Weather Station State

SCAN ACTIVE
ALERT NOT TRIGGERED

Alert Threshold Not Reached Because:

Required convergence was incomplete between:

- organized radar approach,
- persistent/regenerating tower behavior,
- lowering/darkening base,
- coherent local sustainment,
- and direct impact confidence.

Final Summary

The 19341 environment supported:

- intermittent showers,
- fragmented convection,
- and localized precipitation potential.

Regional evidence confirmed:

- active frontal precipitation somewhere within the surrounding mesoscale region,
- but local storm sustainment signals weakened during the observation window.

Operational conclusion:

- Rain became moderately likely.
- Thunder remained conditional.
- Severe potential remained unsupported.
- Scan/watch posture appropriate.
- Alert criteria not met.

Prepared as a structured DI Weather Station operational artifact.

Appendix A — Radar Verification Photo Sequence

Verification note: The late-window radar frames document a narrow precipitation-gradient zone near 19341. The observer point was close enough to the precipitation shield for rain to be more likely than chance, but the strongest embedded return remained displaced south/southeast. This supports the operational classification of SCAN ACTIVE — ALERT NOT TRIGGERED.

Verification Element	Observed Support
Spatial realization	Precipitation reached/approached the 19341 corridor.
Cutoff behavior	Sharp northern edge persisted near the observer point.
Core displacement	Stronger yellow/orange embedded returns remained south/southeast of 19341.

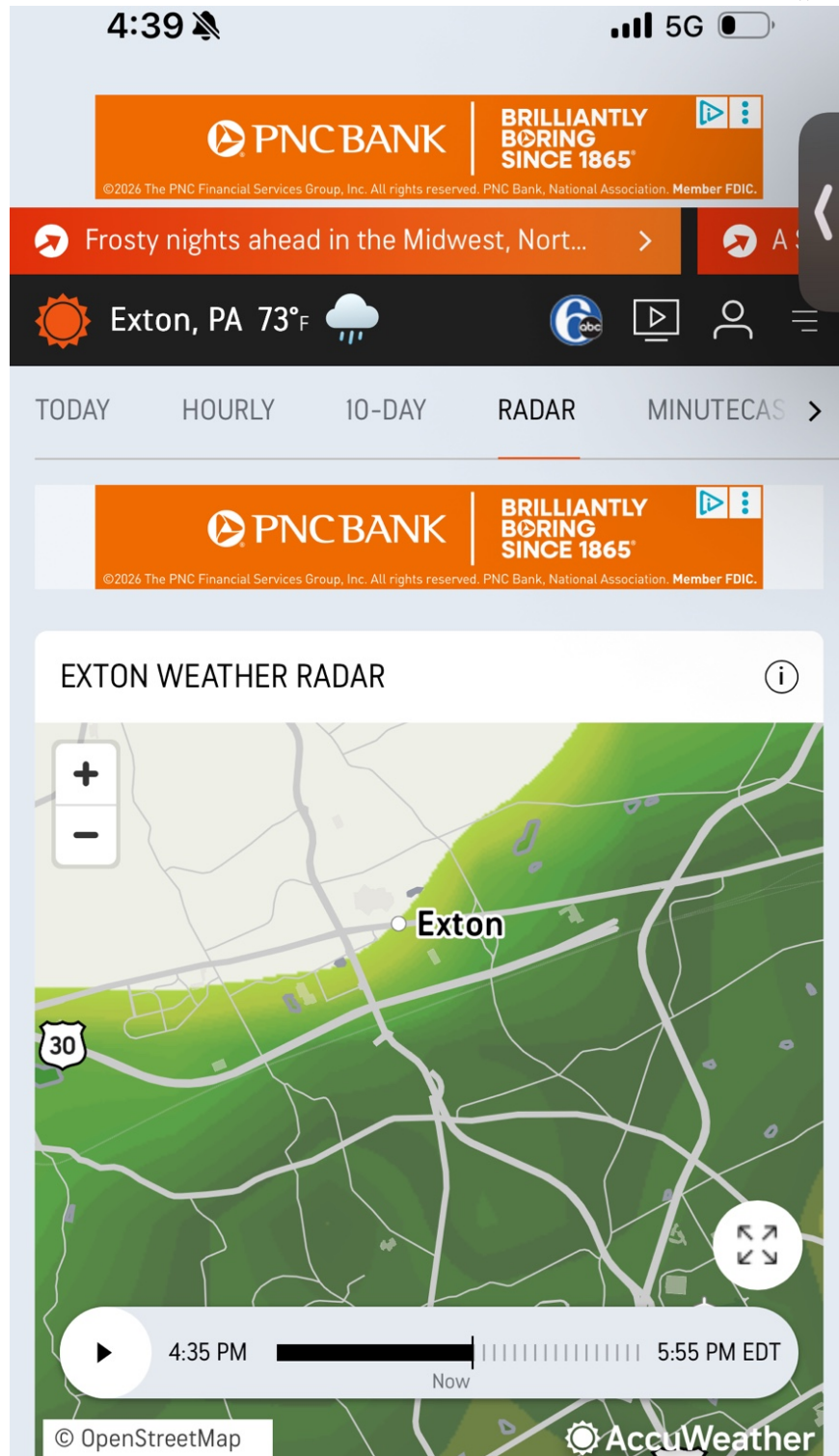


Figure A1 — 4:35 PM radar frame. Light-to-moderate rain returns are near 19341; stronger returns remain displaced south.

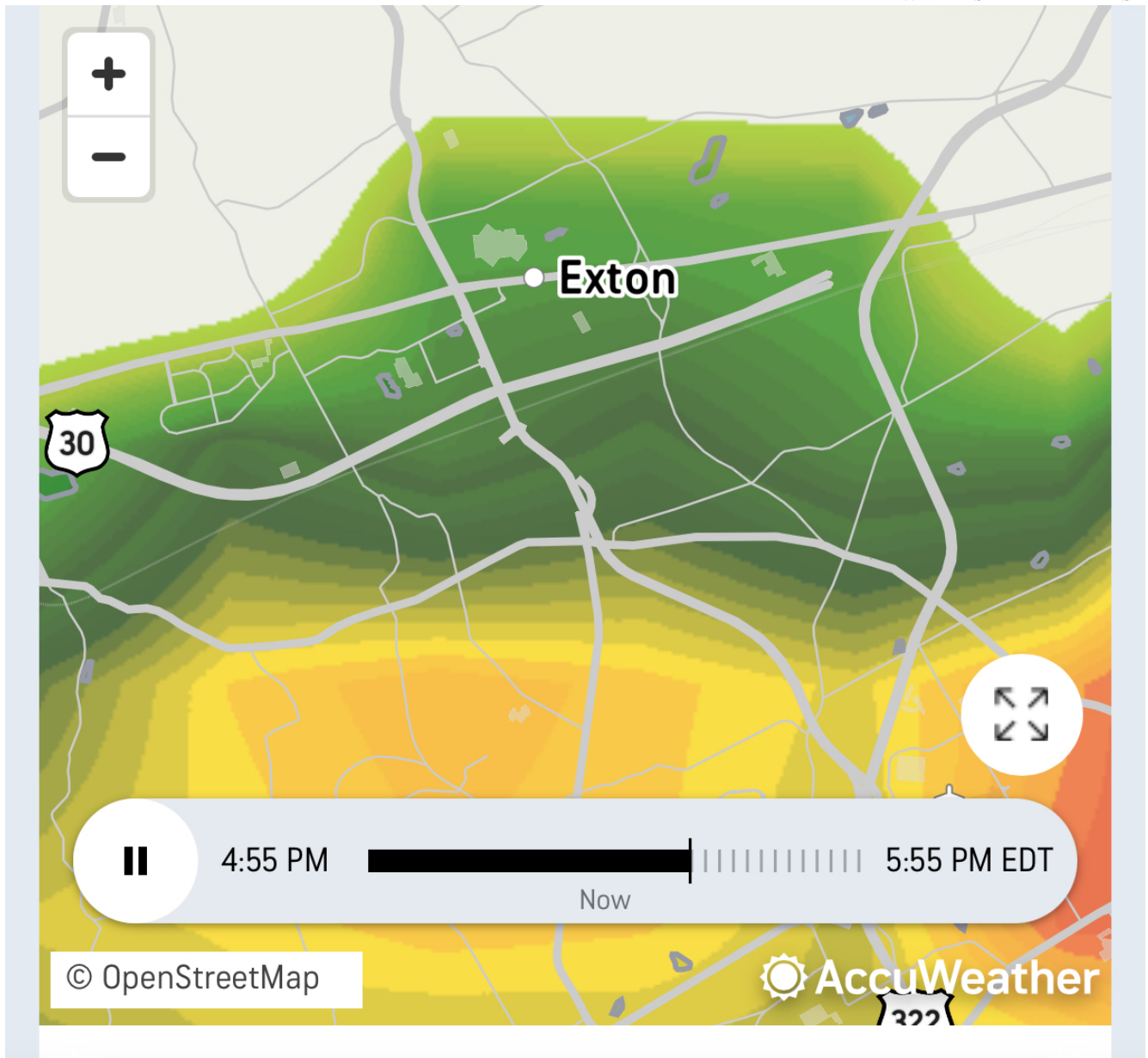


Figure A2 — 4:55 PM radar frame. 19341 remains near the northern gradient; stronger precipitation is south/southeast.